

Sohyla Saeed Abdelrahim

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SUMMARY

Data Scientist specializing in Machine Learning and predictive analytics, with experience building end-to-end AI solutions from data preprocessing to deployment. Passionate about applying data-driven solutions to solve real-world business challenges.

EDUCATION

Bachelor's Degree in Computer Science

Faculty of Computers and Artificial Intelligence, Cairo University

Cumulative GPA: **3.92 / 4.00**

2022 – 2026

GRADUATION PROJECT

Talk2System | *AI-Powered Requirements Engineering Platform*

[GitHub](#) | 2025 – 2026

- Built a full-stack AI-driven platform that automates the software development lifecycle from raw stakeholder conversations to production-ready deliverables.
- Engineered an NLP preprocessing pipeline for transcript cleaning, semantic deduplication, and requirement extraction.
- Designed and trained machine learning classifiers for sentence-level multi-class classification of transcript segments into software engineering categories, achieving **96% accuracy**
- Implemented scalable LLM processing with adaptive chunking and fallback orchestration using **Ollama (Qwen 7B)** and **Google Gemini APIs** to support transcripts exceeding 80K words.
- Developed async background processing, role-aware dashboards, and requirement approval/versioning workflows.
- **Tech stack:** FastAPI, PostgreSQL, scikit-learn, spaCy, FastCoref, Ollama, Gemini API, React

EXPERIENCE

Data Science Trainee

July 2025 – Dec 2025

Digital Egyptian Pioneers Initiative (IBM Curriculum)

- Analyzed and preprocessed large real-world datasets using **Pandas** and **NumPy**, improving data quality by handling missing values and outliers.
- Created dashboards and visualizations using **Matplotlib**, **Seaborn**, and **Streamlit** to communicate insights to non-technical audiences.
- Developed and evaluated 10+ **machine learning** models including classification, clustering, and forecasting systems, improving predictive performance through feature engineering, cross validation and hyperparameter tuning.

PROJECTS

Credit Risk Prediction System (in progress)

[GitHub](#)

- Built an end-to-end credit risk pipeline achieving **0.854 AUC** with XGBoost (Optuna-tuned)
- Tracked model experiments with **MLflow**, enabling reproducible comparison against a logistic regression baseline
- Orchestrated the full pipeline (preprocessing → training) with Apache Airflow DAGs, and architected deployment via FastAPI (serving), Streamlit (UI), and Docker Compose.
- Implemented **Evidently AI** drift monitoring to track production data/model degradation over time.

Sales Forecasting and Optimization (Team Project)

[GitHub](#)

- Built and compared XGBoost, LightGBM, and LSTM forecasting models on retail sales data, achieving **98% accuracy** predictive accuracy.
- Performed EDA, built dashboards, and deployed best performing model using **Docker**.

Material Stream Identification System (Team Project)

[GitHub](#)

- Developed an automated material stream identification system using machine learning techniques.
- Performed data augmentation, feature extraction, and model deployment with **89% accuracy** using **SVM**.

Netflix Show Clustering

[GitHub](#)

- Built a recommendation system using unsupervised learning and **KMeans** clustering.
- Developed a **Streamlit** web app and CLI tools for interactive recommendations.

SMS Spam Classifier

[GitHub](#)

- Designed and deployed an NLP-based spam detection system using TF-IDF and machine learning techniques, achieving **98% accuracy** while reducing false-positive classifications.

CERTIFICATIONS

Associate Data Analyst in SQL – DataCamp

GitHub Foundations – DataCamp

Natural Language Processing in Python – DataCamp

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, JavaScript

Data Science: Pandas, NumPy, Scikit-learn, Feature Engineering, Model Evaluation, Data Cleaning and Preprocessing.

NLP & AI: spaCy, NLTK, LLM Applications, Prompt Engineering.

Visualization: Matplotlib, Seaborn, Streamlit.

Backend & Deployment: FastAPI, PostgreSQL, MySQL, MongoDB, SQL Server, Docker, Git, GitHub.

Concepts: Data Analysis, Statistical Analysis, Machine Learning, Data Warehousing, Managing and Modeling Big Data.

SOFT SKILLS

Problem Solving, Collaboration, Time Management, Adaptability